# Multi-task Hidden Markov Model for Radar Automatic Target Recognition


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**Abstract**

A new radar high-resolution range profile (HRRP) target recognition method is presented via a multi-task learning truncated stick-breaking hidden Markov model (MTL TSB-HMM) with spectrogram feature of HRRP. Experimental results show this method has good recognition and rejection performance.



**Keywords:** Multi-task hidden Markov model (HMM) radar automatic target recognition (RATR).


## 1. Introduction

A high-resolution range profile (HRRP) is the amplitude of coherent summations of the complex time return from target scatterers in each range resolution cell, which contains the target structure signatures, such as target size and scatterer distribution. Therefore, radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1-3, 8].

To make use of the temporal dependence of scattering echoes from a sequence of target aspects, the hidden Markov model (HMM) has been successfully utilized for modelling of sequential-scattering data [1]. However, since the HRRP sample is a typical high-dimensional signal, it is computationally prohibitive to build such a high-dimensional HMM for describing HRRP sequences directly. To avoid this problem, some dimensionality reduction methods are utilized, during which some information of the signal may loss. The work reported here seeks an alternative way of exploiting HMM, in which we characterize the spatial structure across range cells for a single HRRP sample via the hidden Markov structure. Compared with the sequential structure, the spatial structure is much easier to be utilized for practical application.

We utilize multi-task truncated stick-breaking hidden Markov model (MTL TSB-HMM) with spectrogram feature for recognition where the multi-task learning [4] means the multiple tasks of the radar targets are linked and the spectrogram gives a time-frequency representation of HRRP samples. Rather than utilize single-task learning [1] which builds models for each aspect-frame individually (due to target-aspect sensitivity), it is desirable to appropriately share the information among these related data, thus potentially improve recognition performance. The nonparametric form of stick-breaking prior provides automatic model selection. Experimental result shows MTL TSB-HMM obtains good recognition and rejection performance.

## 2. Construction of multi-task truncated hidden Markov model with stick-breaking prior

The spectrogram transform is a common signal processing method which may be represented as:

$$\boldsymbol{Y}(t,f) = \left| \int_{-\infty}^{\infty} x(u) w^*(u-t) e^{-j2\pi f u} du \right|^2 \tag{1}$$

where $x(u)$ is the signal to be transformed and $w^*(*)$ is complex conjugate of the window function.

The advantages of using spectrogram are as follows: (I) It reflects the target physical composition robustly. (II) It contains both physical composition and how the spectrogram density varies with the time. Fig.1 shows the HRRP echoes of the targets and its corresponding spectrogram feature.

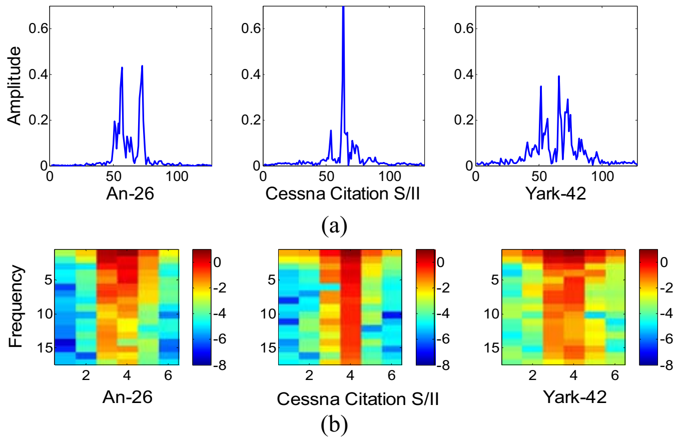


Fig.1 (a) HRRP echoes of three plane targets, (b) Corresponding spectrogram feature of plane HRRP echoes.

The classical hidden Markov model (HMM) [5] has been widely used in target recognition, and it can be modelled as $\mathbf{\Phi} = \{\mathbf{w}_0, \mathbf{W}, \boldsymbol{\theta}\}$ : initial state distribution $\mathbf{w}_0 = \{w_{0,i}\}$ with $w_{0,i} = \Pr(s_1 = i)$, state transition probability $\mathbf{W} = \{w_{i,j}\}$ with $w_{i,j} = \Pr(s_{t+1} = j \mid s_t = i)$, emission likelihood $\boldsymbol{\theta} = \{\theta_i\}$ with $x_t \sim f(\theta_{s_t})$.

In a classical HMM [5], since the number of the states in the HMM is initialized and fixed, we have to specify model structure before learning. To avoid the model selection process, reference [6] proposes the iHMM with stick-breaking priors (SB-HMM), which can develop an HMM with an unknown number of states. In this model, each row of the infinite state transition matrix $\mathbf{W}$ is given a stick-breaking prior. The model is expressed as follows:

$$\mathbf{G}_i = \sum_{j=1}^{\infty} w_{i,j} \delta(\theta_j), \quad w_{i,j} = v_{i,j} \prod_{h=1}^{j-1} (1 - v_{i,h}), \quad v_{i,j} \mid \beta_i \sim \mathbf{Beta}(1, \beta_i)$$

$$\beta_i \sim \mathbf{Ga}(a_\alpha, b_\alpha), \quad \theta_j \sim H; \qquad i = 1, \cdots \infty; \quad j = 1, \cdots, \infty \tag{2}$$

where the mixture distribution $\mathbf{G_i}$ has weights represented as $\mathbf{w}_i = [w_{i,1}, w_{i,2}, ..., w_{i,j}, ...]$, $\delta(\theta_i)$ is a point measure concentrated at $\theta_i$, $\mathbf{Beta}(1, \beta_i)$ represents the Beta distribution with hidden


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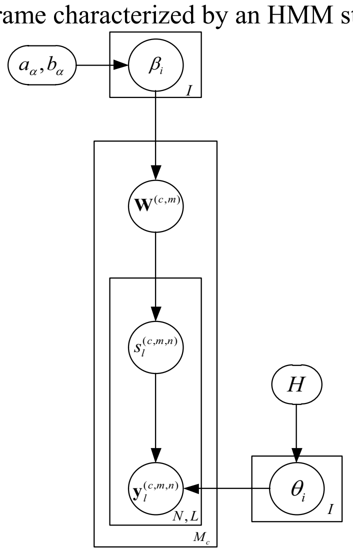
variable $\beta_i$, the drawn variables $\{v_{i,j}\}_{j=1}^{\infty}$ are independent and identically distributed (i.i.d.), $\mathbf{Ga}(a_\alpha, b_\alpha)$ represents the Gamma distribution with preset parameters $a_\alpha, b_\alpha$, and $H$ denotes a prior distribution from which the set $\{\theta_j\}_{j=1}^{\infty}$ is i.i.d. drawn. The initial state probability mass function (pmf) $\mathbf{w}_0$ is also constructed according to an infinite stick-breaking construction. When $\mathbf{w}_i$ terminates at some finite number $I-1$ with $w_{i,I} \equiv 1 - \sum_{j=1}^{I} w_{i,j}$, this result is a draw from a general Dirichlet distribution [7], which is denoted as $\mathbf{w}_i \sim \mathbf{GDD}(\mathbf{1}_{(I-1)\times 1}, [\beta_i]_{(I-1)\times 1})$, where $\mathbf{1}_{(I-1)\times 1}$ denotes an $(I-1)$-length vector of ones, $[\beta_i]_{(I-1)\times 1}$ is an $(I-1)$-length vector of $\beta_i$, and $I$ represents the truncation number of states.

Here, we utilize MTL mechanism for analysing spectrogram features extracted from HRRP data. For a multi-aspect HRRP sequence of target $c$ ($c \in \{1, \cdots, C\}$ with $C$ denoting the number of targets here), we divide the data into $M_c$ aspect frames, e.g. the $m$ th set (here $m \in \{1, \cdots, M_c\}$) is $\{\mathbf{x}^{(c,m,n)}\}_{n=1}^{N}$ where $N$ denotes the number of samples in the frame, and $\mathbf{x}^{(c,m,n)} = [x^{(c,m,n)}(1), \cdots, x^{(c,m,n)}(L_x)]^T$ represents the $n$ th HRRP sample in the $m$ th frame, with $L_x$ denoting the number of range cells in an HRRP sample. Each aspect frame corresponds to a small aspect-sector avoiding scatters' MTRC [2], and the HRRP samples inside each target-aspect frame can be are assumed to be independent and identically distributed (i.i.d). We extract the spectrogram feature of each HRRP sample, and $\mathbf{Y}^{(c,m,n)} = [\mathbf{y}^{(c,m,n)}(1), \cdots, \mathbf{y}^{(c,m,n)}(L_y)]$ denotes the spectrogram feature of $\mathbf{x}^{(c,m,n)}$ as defined in (2) with $L_y$ denoting the number of time bins in spectrogram feature.

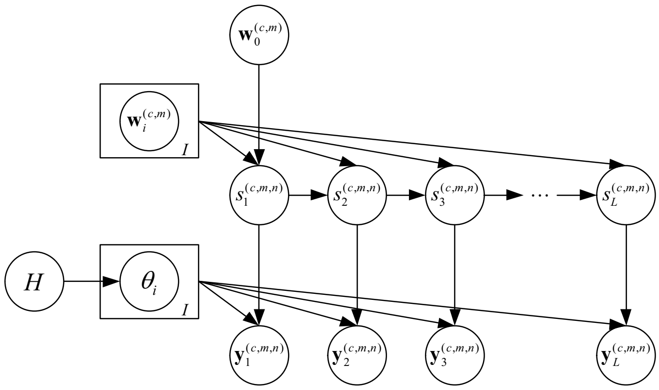
The construction of the MTL TSB-HMM with parameters for target $c$ is represented as:

$$
\begin{aligned}
&\mathbf{y}^{(c,m,n)}(l) \sim f(\boldsymbol{\theta}_{s_l^{(c,m,n)}});\\
&l = 1, \cdots, L_y; \quad n = 1, \cdots, N; \quad m = 1, \cdots, M_c;\\
&s_l^{(c,m,n)} \sim \begin{cases} \mathbf{w}_0^{(c,m)} & if \quad l = 1 \\ \mathbf{w}_{s_{l-1}^{(c,m,n)}}^{(c,m)} & if \quad l \geq 2 \end{cases}\\
&\mathbf{w}_i^{(c,m)} \mid \beta_i \sim \mathbf{GDD}(\mathbf{1}_{(I-1)\times 1}, [\beta_i]_{(I-1)\times 1}),\\
&\beta_i \mid a_\alpha, b_\alpha \sim \mathbf{Ga}(a_\alpha, b_\alpha); \quad i = 0, \cdots, I\\
&\boldsymbol{\theta}_i \sim H; \quad i = 1, \cdots, I
\end{aligned}
\tag{3}
$$

where $\mathbf{y}^{(c,m,n)}(l)$ is $l$ th time chunk of $n$ th sample's spectrogram in the $m$ th aspect-frame of the $c$ th target, $s_l^{(c,m,n)}$ denotes the corresponding state indicator, $a_\alpha, b_\alpha$ are the preset hyperparameters. Here the observation model $f(\cdot)$ is defined as independently normal distribution for computational convenience and each corresponding element in $H(\cdot)$ is normal-gamma distribution to preserve conjugacy requirements. A graphical representation of this model is shown in Fig.2 (a), and Fig.2 (b) depicts that the sequential dependence across time bins of spectrogram feature for a given aspect-frame characterized by an HMM structure.



(a)



(b)

Fig.2 (a) Graph of the MTL TSB-HMM (b) Representation of the HMM structure for the $m$ th aspect frame from target $c$. The spatial dependence across time bins of spectrogram feature for a given aspect frame. For the symbol $L, N, M_c$ or $I$ at the edge of a box indicates that there is a stack of such boxes for the index $l, n, c$ or $i$, where $L$ denotes the number of range cells in an HRRP sample, $N$ denotes the number of samples in an aspect frame, $M_c$ denotes the number of targets, $I$ denotes the truncation number of states.

If learning a separate HMM for each frame of the target, MTL TSB-HMM degenerates into single-task TSB-HMM (STL TSB-HMM). The main difference between MTL TSB-HMM and STL TSB-HMM is: in MTL TSB-HMM, all the multi-aspect frames of one target are learned jointly, each of the $M_c$ tasks of target $c$ is assumed to have an independent state-transition statistics, but the state-dependent observation statistics are shared across these tasks, i.e., the observation parameters are learned via all aspect-frames; while in the STL TSB-HMM, each multi-aspect frame of target $c$ is learned separately, therefore, each target-aspect frame builds its own model and the corresponding parameters are learned just via this aspect-frame.

## 3. Model learning and classification

In this paper, we employ variational Bayesian (VB) inference [6], the goal of Bayesian inference is to estimate the posterior distribution of model parameters $\Phi$. Given the observation data $\mathbf{X}$ and hyper parameters $\boldsymbol{\gamma}$, VB inference is a computational tractable way of computing which seeks a variational distribution $q(\mathbf{\Phi})$ to approximate the true posterior distribution of the latent variables $p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma})$, we obtain the expression:

$$\log p(\mathbf{X}\,|\,\boldsymbol{\gamma}) = L(q(\mathbf{\Phi})) + KL(q(\mathbf{\Phi})\,\|\,p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma})) \quad (4)$$

where $L(q(\mathbf{\Phi})) = \int q(\mathbf{\Phi})\log\frac{p(\mathbf{X}\,|\,\mathbf{\Phi},\boldsymbol{\gamma})p(\mathbf{\Phi}\,|\,\boldsymbol{\gamma})}{q(\mathbf{\Phi})}d\mathbf{\Phi}$ and $KL(q(\mathbf{\Phi})\,\|\,p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma}))$ is the Kullback-Leibler (KL) divergence between the variational distributions $q(\mathbf{\Phi})$ and the true posterior $p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma})$. Since $KL(q(\mathbf{\Phi})\,\|\,p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma}))\geq 0$, and it reaches zero when $q(\mathbf{\Phi}) = p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma})$, this forms a lower bound for $\log p(\mathbf{X}\,|\,\boldsymbol{\gamma})$, so $\log p(\mathbf{X}\,|\,\boldsymbol{\gamma}) \geq L(q(\mathbf{\Phi}))$. The goal of minimizing the KL divergence between the variational distribution and the true posterior is equal to maximize this lower bound,

For the computational convenience and tractability, a factorized $q(\mathbf{\Phi})$ is assumed. i.e. $q(\mathbf{\Phi}) = \prod_k q_k(\varphi_k)$, which has same form as employed in $p(\mathbf{\Phi}\,|\,\mathbf{X},\boldsymbol{\gamma})$. With this assumption, the mean field approximation of the variational distributions for the proposed MTL TSB-HMM with target $c$ may be expressed as:

$$q(\mathbf{\Phi}) = \prod_{m=1}^{M_c}\prod_{i=0}^{I} q(\mathbf{w}_i^{(c,m)})\Big[\prod_{m=1}^{M_c}\prod_{n=1}^{N}\prod_{l=1}^{L_y} q(s_l^{(c,m,n)})\Big]q(\boldsymbol{\theta})q(\boldsymbol{\beta}) \quad (5)$$

where $\{\{\mathbf{w}_i^{(c,m)}\}_{m=1,i=0}^{M_c,I}, \{s_l^{(c,m,n)}\}_{m=1,n=1,l=1}^{M_c,N,L_y}, \boldsymbol{\theta}, \boldsymbol{\beta}\}$ are the latent variables in this MTL model. The detail implementation of MTL TSB-HMM algorithm is follows:

1. Update $q(\boldsymbol{\theta}_i)$:

$$\log q(\theta_i) \propto \log H(\boldsymbol{\theta}_i\,|\,\tilde{\boldsymbol{\theta}}_i) + \sum_{m=1}^{M_c}\sum_{n=1}^{N}\sum_{l=1}^{L_y} < s_{l,i}^{(c,m,n)} > \log f(\mathbf{y}^{(c,m,n)}(l)\,|\,\boldsymbol{\theta}_i);$$
$$i = 1,\cdots I \quad (6)$$

where $\tilde{\boldsymbol{\theta}}_i$ denotes the parameters in the posterior for $H(\cdot)$, $< s_{l,i}^{(c,m,n)} >$ represents the expected number of state indicator $s_l^{(c,m,n)}$ with outcome $i$. If model is conjugate exponential, that is, $H(\cdot)$ is conjugate to the likelihood $f(\cdot)$, we can readily obtain the update equation for $q(\boldsymbol{\theta}_i)$.

2. Update $q(\beta_i)$

We assume $q(\beta_i) = \mathbf{Ga}(\tilde{a}_\alpha^{(i)}, \tilde{b}_\alpha^{(i)})$, the updating equation for $\tilde{a}_\alpha^{(i)}$ and $\tilde{b}_\alpha^{(i)}$ with $m = 1,\cdots,M_c$ are expressed as follows:

$$\begin{cases} \tilde{a}_\alpha^{(i)} = a_\alpha + M_c(I-1) \\ \tilde{b}_\alpha^{(i)} = b_\alpha - \sum_{m=1}^{M_c}\sum_{j=1}^{I-1}[\psi(\tilde{\beta}_{i,j}^{(c,m)}\,|_2) - \psi(\tilde{\beta}_{i,j}^{(c,m)}\,|_1 + \tilde{\beta}_{i,j}^{(c,m)}\,|_2)] \end{cases} ; \quad i = 1,\cdots,J \quad (7)$$

where $\psi(\cdot)$ is the digamma function.

3. Update $q(\mathbf{w}_0^{(c,m)})$ and $q(\mathbf{w}_i^{(c,m)})$

For the given prior $p(\mathbf{w}_0^{(c,m)}) = \mathbf{GDD}(\mathbf{1}_{(I-1)\times 1}, [\beta_0]_{(I-1)\times 1})$ and $p(\mathbf{w}_i^{(c,m)}) = \mathbf{GDD}(\mathbf{1}_{(I-1)\times 1}, [\beta_i]_{(I-1)\times 1})$, with $m = 1,\cdots,M_c$ and $i = 1,\cdots,I$, assume $q(\mathbf{w}_0^{(c,m)}) = \mathbf{GDD}(\tilde{\boldsymbol{\beta}}_0^{(c,m)}\,|_1, \tilde{\boldsymbol{\beta}}_0^{(c,m)}\,|_2)$ and $q(\mathbf{w}_i^{(c,m)}) = \mathbf{GDD}(\tilde{\boldsymbol{\beta}}_i^{(c,m)}\,|_1, \tilde{\boldsymbol{\beta}}_i^{(c,m)}\,|_2)$. The updating equation for $q(\mathbf{w}_0^{(c,m)})$ and $q(\mathbf{w}_i^{(c,m)})$ are given as follows:

$$\begin{cases} \tilde{\beta}_{0,j}^{(c,m)}\,|_1 = 1 + \sum_{n=1}^{N} < s_{1,j}^{(c,m,n)} > \\ \tilde{\beta}_{0,j}^{(c,m)}\,|_2 = \frac{\tilde{a}_\alpha^{(0)}}{\tilde{b}_\alpha^{(0)}} + \sum_{n=1}^{N}\sum_{h=j+1}^{I} < s_{1,h}^{(c,m,n)} > \end{cases} ; j = 1,\cdots,I-1$$

$$\begin{cases} \tilde{\beta}_{i,j}^{(c,m)}\,|_1 = 1 + \sum_{n=1}^{N}\sum_{l=2}^{L_y} < s_{l-1,i}^{(c,m,n)} >< s_{l,j}^{(c,m,n)} > \\ \tilde{\beta}_{i,j}^{(c,m)}\,|_2 = \frac{\tilde{a}_\alpha^{(i)}}{\tilde{b}_\alpha^{(i)}} + \sum_{n=1}^{N}\sum_{h=j+1}^{I}\sum_{l=2}^{L_y} < s_{l-1,i}^{(c,m,n)} >< s_{l,h}^{(c,m,n)} > \end{cases} ; (8)$$
$$i = 1,\cdots,I;\ j = 1,\cdots,I-1$$

4. Update $\log q(s)$

Given the approximate distribution of the other variables, the update equation for $q(\mathbf{s}^{(c,m,n)})$ is given as follows:

$$\log q(\mathbf{s}^{(c,m,n)}) \propto < \log \mathbf{w}_{0,s_1^{(c,m,n)}}^{(c,m)} > + \sum_{l=1}^{L_y-1} < \log \mathbf{w}_{s_l^{(c,m,n)},s_{l+1}^{(c,m,n)}}^{(c,m)} >$$
$$+ \sum_{l=1}^{L_y} < \log f(\mathbf{y}^{(c,m,n)}(l)\,|\,\boldsymbol{\theta}_{s_l^{(c,m,n)}}) >; \quad (9)$$
$$m = 1,\cdots,M_c, \quad n = 1,\cdots,N$$

where $< \cdot >$ denotes the expectation of the associated variables function. One may derive that:

$$< \log \mathbf{w}_{i,j}^{(c,m)} > = \sum_{h=1}^{j-1}([\psi(\tilde{\beta}_{i,h}^{(c,m)}\,|_2) - \psi(\tilde{\beta}_{i,h}^{(c,m)}\,|_1 + \tilde{\beta}_{i,h}^{(c,m)}\,|_2)]$$
$$+ [\psi(\tilde{\beta}_{i,j}^{(c,m)}\,|_1) - \psi(\tilde{\beta}_{i,j}^{(c,m)}\,|_1 + \tilde{\beta}_{i,j}^{(c,m)}\,|_2)]; \quad (10)$$
$$i = 0,\cdots,I, \quad j = 1,\cdots,I$$

In the classification phase, the frame-conditional likelihood of target can be calculated as:

$$p(\mathbf{Y}_{test}^{(c,m)}\,|\,c,m) = \sum_{\mathbf{s}}\left\langle w_{0,s_1^{(c,m)}}^{(c,m)}\right\rangle \prod_{l=2}^{L_y}\left\langle w_{s_{l-1}^{(c,m)},s_l^{(c,m)}}^{(c,m)}\right\rangle \prod_{l=1}^{L_y} f^{(c,m)}(\mathbf{y}_{test}^{(c,m)}(l)\,|\,\left\langle \boldsymbol{\theta}_{s_l^{(c,m)}}\right\rangle)$$

where $\langle\cdot\rangle$ means the posterior expectation for the latent variable over the corresponding distribution on it, e.g., $\left\langle w_{0,s_1^{(c,m)}}^{(c,m)}\right\rangle$ denotes the posterior expectations of initial state probability, $\left\langle w_{s_{l-1}^{(c,m)},s_l^{(c,m)}}^{(c,m)}\right\rangle$ denotes the posterior expectations of state transition probability from state $s_{l-1}^{(c,m)}$ to the $s_l^{(c,m)}$ for the frame $m$ of target $c$, and $\left\langle \boldsymbol{\theta}_{s_l^{(c,m)}}\right\rangle$ denotes the posterior expectations of the observation model parameters associated with state $s_l^{(c,m)}$, with the corresponding state indicator for the $l$ th time chunk $s_l^{(c,m)} \in \{1,\cdots,I\}$. Then $p(\mathbf{Y}_{test}^{(c,m)}\,|\,c,m)$ can be calculated by forward-backward procedure [5] for each $m$ ($m \in \{1,\cdots,M_c\}$) and $c$ ($c \in \{1,\cdots,C\}$).

The testing HRRP sample will be assigned to the class with the maximum class-conditional likelihood, with the assumption that the prior class probabilities are same for all targets of interests: $k = \arg\max_{c,m} (\mathbf{Y}_{test}^{(c)} \mid c, m); \quad c = 1,\cdots,C, \quad m = 1,\cdots,M_c$ (12)

## 4. Experimental results and discussion

The results presented in this paper are based on the measured data, including three actual aircrafts, the radar works on a $C$ band with bandwidth of 400 $MHz$, and the range resolution of the HRRP is about 0.375 $m$. The projections of target trajectories on to the ground plane are displayed in Fig. 3.

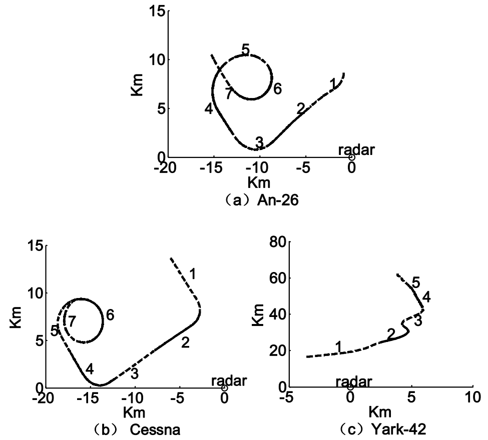



Fig. 3. Projections of target trajectories onto ground. (a) Yak-42; (b) Cessna; (c) An-26.

In the recognition phase, the training data and test data are chosen from different data segments. The 2nd and the 5th segments of Yark-42, the 6th and the 7th segments of Cessna Citation S/II and the 5th and the 6th segments of An-26 are taken as the training samples while the remaining data are left for testing. The training data are divided into equal length frames. In training phase, there are 50 frames for An-26, 50 frames for Cessna and 35 frames for Yark-42. The pre-process of the data is the same as ref. [3]. We choose the window length of 33 range cells and overlap length of 16 range cells for spectrogram feature extraction. In Table. 1, we present the confusion matrices and average recognition rates of the STL TSB-HMM and MTL TSB-HMM. The average recognition rate obtained by MTL TSB-HMM is gained about 2.3%.

Table.1 Confusion Matrices of Three Planes using Single-task Learning and Multi-task Learning (in percentage)

| | **Single-task learning** | | | **Multi-task learning** | | |
|---|---|---|---|---|---|---|
| | An-26 | Cessna | Yark-42 | An-26 | Cessna | Yark-42 |
| An-26 | 93.8 | 0.7 | 2.5 | 95.8 | 0.4 | 1.9 |
| Cessna | 4.4 | 88.3 | 1.1 | 3.5 | 92.3 | 0.8 |
| Yark-42 | 1.8 | 11.0 | 96.4 | 0.7 | 7.3 | 97.3 |
| Average | 92.8 | | | 95.1 | | |

The ability to reject unseen confuser targets is also an important indicator of model performance. So we also perform a rejection experiment. 18000 truck HRRPs generated by the simulation software XPATCH are used as confuser target. The receiver operating characteristic (ROC) curves of MTL SB-HMM are shown in Fig. 4. The area under curve (AUC) of MTL SB-HMM is 90.5%.

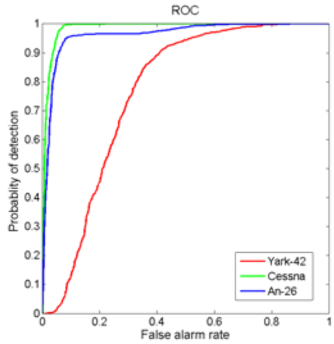



Fig.4 ROC curves of three plane targets for the MTL SB-HMM.

## 4. Conclusion

We have presented a multi-task learning truncated stick-breaking hidden Markov model (MTL TSB-HMM) with spectrogram feature for radar HRRP target recognition. The construction of this model allows VB inference, which can learn parameters of MTL TSB-HMM very fast. The experimental results show that MTL TSB-HMM using spectrogram feature have better recognition rate than STL TSB-HMM, and MTL TSB-HMM have good rejection performance.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